
A R C H I T E C T U R E A D D E N D U M — A C P v 2 . 0

Multi-LLM Orchestration & MAPE-K Autonomic Engine

*Choose the right intelligence for every phase of
autonomous cloud management and intelligent digital systems engineering*

Infrastructure · Containers · Data Analytics · AI/ML · Gen AI / Agentic AI

AWS · Azure · GCP · NIST SP 800-53 Rev 5 · FISMA Moderate



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1. Overview — Why Multi-LLM MAPE-K?

ACP v2.0 introduces a fundamental enhancement to the platform's autonomic intelligence layer: the ability to assign different Large Language Models to different phases of the MAPE-K control loop, and to different workload types, giving operators precise control over the trade-off between reasoning quality, cost, latency, and compliance boundary.

The MAPE-K (Monitor · Analyze · Plan · Execute · Knowledge) framework is the gold standard architecture for autonomous computing systems, originally formalized by IBM Research. ACP implements MAPE-K as a continuous, AI-powered closed loop that governs all five workload types — infrastructure, containerized applications, data analytics, AI/ML, and generative / agentic AI solutions — across AWS, Azure, and GCP simultaneously.

"Not every autonomic decision requires the same intelligence. A fast, cheap model handles routine monitoring narration. A powerful reasoning model handles novel threat analysis. ACP routes each task to the right LLM — automatically."

Key principle: The ACP LLM Router normalizes all provider APIs behind a single interface. Switching LLMs for any MAPE-K phase requires no code changes and no platform redeployment — just a configuration update in the ACP portal.

2. Multi-LLM MAPE-K Architecture Diagram

The diagram below shows the complete architecture: the 8 supported LLM providers, the ACP LLM Router, the 5 MAPE-K phases with per-phase LLM assignment, the 5 workload types, the 3 cloud platform targets, and the Knowledge Base that grounds every LLM decision in audited, persistent context.



Figure 4 — ACP Multi-LLM Orchestration & MAPE-K Autonomic Engine. LLM providers (top) → Router → 5 MAPE-K phases (center) → 5 workload types → 3 cloud targets → Knowledge Base (bottom).

3. Supported LLM Providers

ACP supports eight LLM provider ecosystems through a unified API adapter layer. All providers are accessed through the ACP LLM Router, which normalizes request formats, manages authentication via HashiCorp Vault, enforces token budgets, and logs all interactions for audit compliance.

LLM Provider	Models Available	API Endpoint	Primary Strength	Best MAPE-K Phase	Notes
Anthropic Claude	claude-opus-4 claude-sonnet-4 claude-haiku-4	Anthropic API AWS Bedrock	Reasoning, Code, Compliance analysis	Analyze · Plan · Execute	Hot-swappable via LLM Router. No redeployment required.
OpenAI	gpt-4o / gpt-4o-mini o1 / o1-mini o3 / o3-mini	OpenAI API Azure OpenAI	General reasoning, Function calling, Code	All phases	Hot-swappable via LLM Router. No redeployment required.
Google Gemini	gemini-2.0-flash gemini-2.5-pro gemini-ultra	Vertex AI Google AI Studio	Long context, Multimodal, Data analysis	Monitor · Analyze · Plan	Hot-swappable via LLM Router. No redeployment required.
Meta Llama	Llama 3.3 70B Llama 3.3 405B Code Llama	Bedrock Vertex AI Self-hosted	Open weights, Privacy, On-prem deployment	Execute · Knowledge	Hot-swappable via LLM Router. No redeployment required.
AWS Bedrock	Claude · Titan Nova · Llama Mistral on Bedrock	AWS Bedrock Converse API	Multi-model gateway, AWS-native integration	All phases (AWS)	Hot-swappable via LLM Router. No redeployment required.
Azure OpenAI	GPT-4o deployments o1 deployments Phi-4 / SLMs	Azure OpenAI Service endpoint	Azure-native, Data residency, SLAs	All phases (Azure)	Hot-swappable via LLM Router. No redeployment required.

All LLM provider API keys are stored in HashiCorp Vault and synchronized to cloud-native secret stores (AWS Secrets Manager, Azure Key Vault, GCP Secret Manager). Keys are rotated on the standard 90-day schedule via ACP's secret rotation orchestration — never stored in environment variables or source code.

4. MAPE-K Autonomic Control Loop

MAPE-K is the architectural backbone of ACP's autonomous operation. Each phase has a distinct role, a distinct computational profile (latency tolerance, reasoning depth required, output type), and therefore benefits from a different LLM optimized for that profile. The following table defines each phase's function, recommended LLMs, and key signals or outputs.

Phase	Name & Role	LLM Function	Recommended LLMs	Key Signals / Outputs	Description
M	Monitor Observe · Collect · Normalize	Continuously ingests telemetry from all clouds and workloads. Normalizes CloudWatch / Azure Monitor / Cloud Monitoring events into a unified structured feed. LLM narrates anomalies and summarizes signal volumes.	Claude Haiku · Gemini Flash · GPT-4o-mini · Phi-4 (low-cost, high-throughput)	OTel metrics/logs/traces · SecurityHub / Defender / SCC findings · Container health · Cost anomalies · ML drift metrics · LLM token burn rate	See diagram for full signal list.
A	Analyze Reason · Classify · Diagnose	LLM performs deep reasoning over full event context to classify severity, identify root cause, assess blast radius across workloads, map threats to MITRE ATT&CK, and correlate compliance drift.	Claude Opus · o1 / o3 · Gemini 2.5 Pro · Llama 405B (high reasoning, complex multi-step)	Multi-signal correlation · MITRE ATT&CK threat mapping · ML degradation diagnosis · Agentic AI loop failure · Cost spike attribution · Cross-service dependency graph	See diagram for full signal list.
P	Plan Strategy · Optimize · Validate	LLM generates ranked remediation strategies, evaluates trade-offs (cost / speed / risk), selects optimal action path, and produces detailed execution plan with rollback checkpoints and compliance pre-validation.	Claude Sonnet · Gemini 2.5 Pro · o1 · GPT-4o · Llama 405B (balanced reasoning + output)	Ranked remediation options with risk/impact/cost · Terraform change plan · Scaling strategy selection · Cost optimization plan · ML retraining plan · Multi-cloud placement decision	See diagram for full signal list.
E	Execute Generate · Deploy · Remediate	LLM generates exact artifacts — Terraform HCL, Helm values, pipeline YAML, API payloads, MLOps pipeline configs, agent tool definitions. Execution Engine dispatches to cloud APIs with full audit trail.	Claude Sonnet · GPT-4o · Gemini Flash · Code Llama (high output quality, code-specialized)	TF HCL generation · Kubernetes manifests · GitHub Actions trigger · SageMaker/Vertex/Azure ML pipeline · Agentic AI tool code · Data pipeline DAGs · Auto-	See diagram for full signal list.

Phase	Name & Role	LLM Function	Recommended LLMs	Key Signals / Outputs	Description
K	Knowledge Learn · Retain · Evolve	Every MAPE-K cycle outcome — decision made, action executed, result verified, cost incurred — is written to the immutable Knowledge Base. All future LLM reasoning is grounded in this persistent, audited context.	Claude Haiku · Gemini Flash · GPT-4o-mini · Llama 70B (efficient summarization, distillation)	remediation API calls Runbook library · Performance & cost time-series · Compliance audit trail · Threat intelligence · ML performance history · LLM model performance benchmarks per task type	See diagram for full signal list.

4.1 The Knowledge Base — Persistent Context for All LLM Decisions

The Knowledge Base is the 'K' in MAPE-K and the most critical architectural component for long-term autonomic intelligence. Without it, each LLM invocation starts from zero context. With it, every new analysis is grounded in the full history of the system's behavior, prior incidents, proven remediations, and cost outcomes.

- **Runbook Library:** Indexed by workload type and failure pattern. Every successful Execute phase outcome adds a new proven procedure.
- **Performance & Cost History:** Resource utilization, cost per workload, model inference latency and cost, scaling decision outcomes — all timestamped.
- **Compliance & Threat Intelligence:** Full NIST control audit trail, threat incident history, SBOM-based CVE cross-references.
- **LLM Performance Log:** Tracks latency, accuracy, token cost, and hallucination rate per model per task type. Feeds the Router's model selection algorithm for continuous optimization.

The Knowledge Base is stored in immutable, WORM-protected storage (S3 / Azure Blob / GCS) with 7-year retention — satisfying NIST AU-9 and AU-11 audit retention requirements. Every LLM prompt and response is included in the audit record.

5. ACP LLM Router & Orchestration Layer

The LLM Router is the central nervous system of ACP's multi-LLM architecture. It sits between all MAPE-K phase executors and all LLM provider APIs, normalizing request formats, enforcing routing rules, managing budgets, and capturing telemetry for every interaction.

Router Capability	What It Does	Operational Detail
Per-Phase LLM Assignment	Configure a different LLM model for each of the 5 MAPE-K phases independently	Use a fast/cheap model for Monitor, a powerful reasoning model for Analyze, a code-specialized model for Execute
Per-Workload-Type Override	Override the phase-level LLM selection for specific workload types	e.g., use Codestral specifically for Container workload Execute phase, Claude Opus for Gen AI Analyze phase
Fallback Chain on Failure	Define an ordered fallback chain: if primary LLM API fails, auto-route to next	Primary: Claude Sonnet → Fallback 1: GPT-4o → Fallback 2: Gemini Flash. Zero disruption to autonomic loop
Cost-Aware Routing	Route to cheaper model when task complexity is below a defined threshold	Simple log summarization → Haiku/Flash. Complex root cause across 50 signals → Opus/o1. Token budget enforced per phase
Latency-Based Selection	For time-critical phases (Execute), select model by measured P95 latency	During incident response, prefer lowest-latency model meeting quality bar. Latency tracked in Knowledge Base
Data Residency Enforcement	Enforce that LLM API calls stay within a geographic boundary or cloud provider	EU workloads must use Azure OpenAI EU endpoint. US GovCloud must use Bedrock us-gov-east-1. Enforced via routing rules
Token Budget Management	Set hard token limits per MAPE-K phase and per workload type per cycle	Prevents runaway LLM cost during high-frequency monitoring. Budget resets per cycle. Alerts on budget consumption
Prompt Versioning & Audit	Every prompt sent to every LLM is versioned, stored, and included in audit log	Full prompt + response audit trail for NIST AU-2/AU-12 compliance. Enables prompt regression testing and rollback
A/B Model Evaluation	Run two LLMs in parallel for the same task and compare outputs before executing	Shadow mode: new model generates plan alongside production model. Confidence score compared before routing to Execute

Router Capability	What It Does	Operational Detail
Hot-Swap Without Redeployment	Change LLM assignments for any phase via ACP portal — no pipeline restart needed	Operations team can instantly switch from Claude to GPT-4o for Analyze phase via a dropdown — effective immediately
Usage Telemetry Per Model	Track tokens consumed, latency, cost, and success rate per model per phase	Feeds the Knowledge Base LLM Performance Log. Enables data-driven model selection and cost forecasting
Provider API Key Management	All LLM provider API keys stored in HashiCorp Vault and synced to cloud secrets	Keys rotated on schedule via ACP secret rotation orchestration. Never stored in environment variables or code
✦ Unified Mode (Simple)	Configure one LLM for ALL MAPE-K phases with a single selection	Ideal for getting started or for teams that want simplicity. One dropdown in the ACP portal sets the same model across Monitor, Analyze, Plan, Execute, and Knowledge simultaneously. Can be upgraded to per-phase assignment at any time without data migration.

6. MAPE-K Applied Across All Five Workload Types

ACP applies the full MAPE-K loop to each of five distinct workload types. Each type has its own monitoring signals, analysis patterns, planning strategies, execution mechanisms, and knowledge retention requirements — and each can be independently configured with its own LLM assignment per phase.

Workload	Name	Description	Technology Stack	MAPE-K Application
	Infrastructure & IaC	Multi-cloud resource provisioning, networking, security posture, cost management — continuously governed by autonomous MAPE-K loops.	Terraform · Terra grunt · OPA/Rego AWS: ECS · RDS · VPC · WAF Azure: AKS · Azure DB · VNet GCP: GKE · Cloud SQL · Cloud Armor	M: Cost anomaly + Config drift · A: Root cause classification · P: IaC remediation plan · E: Terraform Apply · K: Config baseline history
	Containerized Apps	Node.js / Python / Go / Java microservices on ECS Fargate, AKS, and GKE Autopilot — autonomous health, scaling, deployment, and supply chain security.	Docker multi-stage · Cosign signing Syft SBOM · Grype CVE scanning Helm · kubectrl · Blue-Green deploy HPA / VPA auto-scaling	M: Health + Latency + Error rate · A: Failure root cause · P: Scale / rollback strategy · E: Helm upgrade / ECS update · K: Deployment outcome history
	Data & Analytics	Streaming ingestion, batch processing, data warehousing, and orchestrated pipelines — autonomic data quality, pipeline SLA, and cost management.	Ingest: Kinesis / Event Hubs / Pub/Sub Process: EMR / Databricks / Dataproc Store: S3 / ADLS Gen2 / BigQuery Orchestrate: Airflow / Composer / MWAA	M: Pipeline SLA + Data quality · A: Data quality root cause · P: Partition / index strategy · E: DAG update / cluster resize · K: Pipeline performance history
	AI / ML Workloads	Full MLOps lifecycle — training, model registry, inference serving, feature stores, continuous drift detection, and automated retraining pipelines.	Train: SageMaker / Azure ML / Vertex AI Serve: Managed endpoints (all 3 clouds) Features: Feature Store (all 3 clouds) Monitor: Model drift · Data drift · Bias	M: Accuracy + Latency + Drift · A: Degradation cause (data / concept / model) · P: Retrain strategy · E: MLOps pipeline trigger · K: Model performance history
	Gen AI & Agentic AI	LLM-powered applications, RAG pipelines, multi-agent orchestration systems — autonomic cost, latency, hallucination, and	LLM APIs: Bedrock / Azure OAI / Vertex Vector: OpenSearch / AI Search / AlloyDB	M: Token cost + Latency + Hallucination · A: Agent loop failure / RAG quality · P: Model swap / guardrail update · E: Config deploy / router update · K: LLM perf benchmark log

Workload	Name	Description	Technology Stack	MAPE-K Application
		reliability management for AI-native solutions.	Agents: LangGraph / AutoGen / CrewAI Guardrails: Bedrock GR / Azure AI Content	

6.1 Generative AI & Agentic AI — Special Considerations

Gen AI and Agentic AI workloads are unique because ACP itself uses LLMs as part of the MAPE-K loop — meaning the platform is autonomically managing other LLM-powered systems. This creates a meta-layer of intelligence that requires special attention:

- **Token cost as a first-class metric:** Monitor phase tracks token consumption, cost per request, and burn rate in real time. Cost anomalies trigger Analyze phase.
- **Hallucination and quality monitoring:** Integration with LLM observability tools (LangSmith, Arize, Weights & Biases) feeds hallucination rate and answer quality scores into the Monitor signal stream.
- **Agent loop failure detection:** Infinite loops, tool call failures, context window exhaustion, and guardrail violations are detected by Monitor and classified by Analyze.
- **Model swap without downtime:** When Analyze determines a model is underperforming, Plan generates a router configuration change, Execute deploys it, and the running application transparently switches to the better model.
- **Multi-cloud LLM placement:** Plan phase can recommend routing inference traffic between Bedrock, Azure OpenAI, and Vertex AI based on live pricing, latency, and quota availability.

7. Compliance Across the MAPE-K Loop

Every MAPE-K phase operation is governed by NIST SP 800-53 Rev 5 controls. The autonomic loop does not bypass compliance — it enforces it continuously and automatically.

- **AU-2 / AU-12 (Audit):** Every LLM prompt, response, decision, and executed action is written to the immutable audit log. MAPE-K cycle records include the full reasoning chain.
- **CA-7 (Continuous Monitoring):** The Monitor phase IS the CA-7 control — automated, continuous, across all clouds and workload types simultaneously.
- **CM-2 / CM-6 (Configuration Management):** All Terraform changes generated by the Execute phase pass through the OPA compliance gate before application.
- **IR-4 / IR-6 (Incident Response):** The Analyze → Plan → Execute chain constitutes an automated IR capability for HIGH/CRITICAL findings, with PagerDuty notification and forensic state capture.
- **RA-5 (Vulnerability Monitoring):** SBOM-based CVE cross-referencing is a Monitor phase signal. Zero-day response is a pre-defined Execute phase runbook.
- **SI-4 (System Monitoring):** The full MAPE-K Monitor phase covers SI-4 across all clouds, workloads, and the LLM layer itself.
- **SA-10 (Developer Configuration Management):** All code and IaC generated by the Execute phase is signed (Cosign), versioned, and included in the SBOM artifact store.

8. Conclusion

The Multi-LLM MAPE-K architecture transforms ACP from a platform that uses AI for code generation into a platform that uses AI for continuous autonomous governance of the entire cloud estate. Different tasks demand different intelligence — the Monitor phase needs speed, the Analyze phase needs deep reasoning, the Execute phase needs precise code generation, and the Knowledge phase needs efficient summarization.

By making every MAPE-K phase independently configurable with any of eight LLM providers, and by applying this framework across all five workload types on all three clouds, ACP delivers on the original promise of autonomic computing: a system that manages itself with the same intelligence, adaptability, and tirelessness as the human autonomic nervous system.

"Every cloud, every workload, every phase — the right intelligence, applied continuously, governed completely."

For architectural reviews, pilot program onboarding, or technical deep-dives on specific workload type integrations, please contact the Autonomic Solutions LLC engineering team.